HOMEWORK 3: SOLUTIONS

1. (Divisibility) Prove that

$$1+\sqrt{-2} \mid a+b\sqrt{-2}$$

in $\mathbb{Z}[\sqrt{-2}]$ if and only if $a \equiv b \pmod 3$.

Solution. Observe that $1+\sqrt{-2} \mid a+b\sqrt{-2}$ if and only if

$$\frac{a+b\sqrt{-2}}{1+\sqrt{-2}} = \frac{(a+b\sqrt{-2})(1-\sqrt{-2})}{(1+\sqrt{-2})(1-\sqrt{-2})} = \frac{(a+2b)+(b-a)\sqrt{-2}}{3} \in \mathbb{Z}[\sqrt{-2}],$$

which is true if and only if $a+2b \equiv 0 \pmod 3$ and $b-a \equiv 0 \pmod 3$. These congruence conditions are equivalent to $a \equiv b \pmod 3$.

2. (Norm and conjugates) Let $d \in \mathbb{Z}$ and $m \geq 1$ be integers. For any $\alpha \in \mathbb{Z}[\sqrt{d}]$, a *multiplicative inverse* of α modulo m is an element $\beta \in \mathbb{Z}[\sqrt{d}]$ such that $m \mid (\alpha\beta - 1)$.

(1) Prove that α has a multiplicative inverse modulo m if and only if $\gcd(N(\alpha), m) = 1$.

Solution. If $\gcd(N(\alpha), m) = 1$, then there is some $c \in \mathbb{Z}$ such that $cN(\alpha) \equiv 1 \pmod m$. Since $N(\alpha) = \alpha\overline{\alpha}$, this implies

$$m \mid cN(\alpha) - 1 = c\alpha\overline{\alpha} - 1 = \alpha\beta - 1,$$

where $\beta = c\overline{\alpha} \in \mathbb{Z}[\sqrt{d}]$. Thus, α has a multiplicative inverse modulo m.

Conversely, if α has a multiplicative inverse modulo m, say $\beta \in \mathbb{Z}[\sqrt{d}]$, then $m \mid \alpha\beta - 1$. In other words, there is some $\gamma \in \mathbb{Z}[\sqrt{d}]$ such that $\alpha\beta = 1 + m\gamma$. Taking the norms, we get

$$N(\alpha)N(\beta) = N(\alpha\beta) = N(1+m\gamma) = (1+m\gamma)(1+m\overline{\gamma}) = 1 + m(\gamma + \overline{\gamma} + mN(\gamma)).$$

It follows that $\gcd(N(\alpha), m) \mid 1$ in $\mathbb{Z}[\sqrt{d}]$, so $\gcd(N(\alpha), m) = 1$ in $\mathbb{Z}[\sqrt{d}]$. In particular, $\gcd(N(\alpha), m) = 1$ in $\mathbb{Z}$. Alternatively, if we write $\gamma = x + y\sqrt{d}$ with $x, y \in \mathbb{Z}$, then

$$N(\alpha)N(\beta) = N(\alpha\beta) = N(1+m\gamma) = (1+mx)^2 - dm^2y^2 \equiv 1 \pmod m.$$

This also implies that $\gcd(N(\alpha), m) = 1$.

(2) Show that if $c \in \mathbb{Z}$ satisfies $cN(\alpha) \equiv 1 \pmod m$, then $c\overline{\alpha}$ is a multiplicative inverse of α modulo m.

Solution. If $cN(\alpha) \equiv 1 \pmod m$, then

$$m \mid cN(\alpha) - 1 = c\alpha\overline{\alpha} - 1 = \alpha\beta - 1,$$

with $\beta = c\overline{\alpha} \in \mathbb{Z}[\sqrt{d}]$. Thus, β is a multiplicative inverse of α modulo m.

(3) In $\mathbb{Z}[\sqrt{5}]$, find a multiplicative inverse of $3+\sqrt{5}$ modulo 7.

Solution. Since $N(3+\sqrt{5}) = 3^2 - 5 \cdot 1^2 = 4$ and $2N(3+\sqrt{5}) \equiv 1 \pmod 7$, part (2) gives the multiplicative inverse $2(3-\sqrt{5}) = 6 - 2\sqrt{5}$ of $3+\sqrt{5}$ modulo 7.

3. (Euclidean algorithm in $\mathbb{Z}[i]$) Use the Euclidean algorithm in $\mathbb{Z}[i]$ to find

$$\gcd(5+3i, 3+11i).$$

Then express a gcd as a linear combination of $5+3i$ and $3+11i$.

Solution. Since $N(5+3i) = 34 < N(3+11i) = 130$, we divide $3+11i$ by $5+3i$ using the division algorithm in $\mathbb{Z}[i]$:

$$\frac{3+11i}{5+3i} = \frac{(3+11i)(5-3i)}{(5+3i)(5-3i)} = \frac{48+46i}{34} = \frac{24}{17} + \frac{23}{17}i.$$

Choose integers $x = 1$ closest to $24/17$ and $y = 1$ closest to $23/17$. Then the remainder is

$$3+11i - (1+i)(5+3i) = 3+11i - (2+8i) = 1+3i.$$

Similarly, we divide $5+3i$ by $1+3i$:

$$\frac{5+3i}{1+3i} = \frac{(5+3i)(1-3i)}{(1+3i)(1-3i)} = \frac{14-12i}{10} = \frac{7}{5} - \frac{6}{5}i.$$

Choose integers $x = 1$ closest to $7/5$ and $y = -1$ closest to $-6/5$. Then the remainder is

$$5+3i - (1-i)(1+3i) = 5+3i - (4+2i) = 1+i.$$

Repeating, we divide $1+3i$ by $1+i$:

$$\frac{1+3i}{1+i} = \frac{(1+3i)(1-i)}{(1+i)(1-i)} = \frac{4+2i}{2} = 2+i \in \mathbb{Z}[i],$$

so $1+i \mid 1+3i$. Hence, $\gcd(5+3i, 3+11i) = 1+i$. We also have

$$\begin{aligned} 3+11i &= (1+i)(5+3i) + 1 + 3i \\ 5+3i &= (1-i)(1+3i) + 1 + i \\ 1+3i &= (2+i)(1-i). \end{aligned}$$

Backward substitution gives

$$\begin{aligned} 1+i = 5+3i - (1-i)(1+3i) &= 5+3i - (1-i)[3+11i - (1+i)(5+3i)] \\ &= [1+(1-i)(1+i)](5+3i) - (1-i)(3+11i) \\ &= 3(5+3i) - (1-i)(3+11i). \end{aligned}$$

4. (Prime factorization in $\mathbb{Z}[\sqrt{-2}]$) Factor 165 into primes in $\mathbb{Z}[\sqrt{-2}]$. In particular, explain why every factor in your factorization is a prime.

Solution. We first factor 165 as $165 = 3 \cdot 5 \cdot 11$. Since $3, 11 \equiv 3 \pmod 8$ whereas $5 \equiv 5 \pmod 8$, the prime number 5 remains prime in $\mathbb{Z}[\sqrt{-2}]$, but the prime numbers 3 and 11 can be further factored in $\mathbb{Z}[\sqrt{-2}]$:

$$3 = (1+\sqrt{-2})(1-\sqrt{-2}), \qquad 11 = (3+\sqrt{-2})(3-\sqrt{-2}).$$

Hence,

$$165 = (1+\sqrt{-2})(1-\sqrt{-2}) \cdot 5 \cdot (3+\sqrt{-2})(3-\sqrt{-2}),$$

where each factor is a prime in $\mathbb{Z}[\sqrt{-2}]$.

5. (Nonunique factorization in $\mathbb{Z}[\sqrt{d}]$) Suppose that $d \in \mathbb{Z}$ is not a perfect square. We show that $\mathbb{Z}[\sqrt{d}]$ is not a UFD for infinitely many positive and negative values of d.

(1) Prove that $2 \in \mathbb{Z}[\sqrt{d}]$ is a prime whenever $d \leq -3$.

Solution. Suppose $\alpha = x + y\sqrt{d} \mid 2$ is neither a unit or an associate of 2. Since $0 < N(\alpha) = x^2 - dy^2 \mid N(2) = 4$, we have $N(\alpha) = x^2 - dy^2 = 2$. Given that $d \leq -3$, we must have $y = 0$ and so $x^2 = 2$, which is impossible for $x \in \mathbb{Z}$. Hence, 2 is a prime in $\mathbb{Z}[\sqrt{d}]$.

(2) Suppose that $d > 0$ is divisible by a prime number $p \equiv 5 \pmod 8$. Prove that $2 \in \mathbb{Z}[\sqrt{d}]$ is again a prime.

Solution. Again, suppose $\alpha = x + y\sqrt{d} \mid 2$ is neither a unit or an associate of 2. Then $N(\alpha) = x^2 - dy^2 \mid N(2) = 4$ implies $N(\alpha) = x^2 - dy^2 = \pm 2$. Since d is divisible by a prime number $p \equiv 5 \pmod 8$, we have $x^2 \equiv \pm 2 \pmod p$, which means that $2 \text{ R } p$ or $-2 \text{ R } p$. But neither is true, since

$$\left(\frac{\pm 2}{p}\right) = \left(\frac{\pm 1}{p}\right)\left(\frac{2}{p}\right) = \left(\frac{2}{p}\right) = -1$$

for $p \equiv 5 \pmod 8$, meaning that $\pm 2 \text{ N } p$. Hence, 2 is a prime in $\mathbb{Z}[\sqrt{d}]$.

(3) Suppose either that $d \leq -3$ or that d is divisible by a prime number $p \equiv 5 \pmod 8$. Show that $\mathbb{Z}[\sqrt{d}]$ is not a UFD by considering the factorizations

$$\begin{aligned} d &= 2 \cdot \frac{d}{2} = \sqrt{d} \cdot \sqrt{d}, && \text{when } 2 \mid d, \\ 1 - d &= 2 \cdot \frac{1-d}{2} = (1 + \sqrt{d})(1 - \sqrt{d}), && \text{when } 2 \nmid d. \end{aligned}$$

Consequently, the only negative values of d for which $\mathbb{Z}[\sqrt{d}]$ is a UFD are $d = -1, -2$ discussed in class.

Solution. Parts (1) and (2) show that in either case, 2 is prime. If $\mathbb{Z}[\sqrt{d}]$ were a UFD, then the above two factorizations of d and $1 - d$ would imply that $2 \mid \sqrt{d}$ when $2 \mid d$, and $2 \mid 1 + \sqrt{d}$ or $2 \mid 1 - \sqrt{d}$ when $2 \nmid d$. But none of these can occur, because the assumption that d is not a square implies $\sqrt{d} \notin \mathbb{Q}$ and thus

$$\frac{\sqrt{d}}{2}, \frac{1 \pm \sqrt{d}}{2} \notin \mathbb{Z}[\sqrt{d}].$$

6. (Units and Pell's equations) Use the continued fraction expansion of $\sqrt{14}$ to find all the units in $\mathbb{Z}[\sqrt{14}]$. Does any unit provide a solution to $x^2 - 14y^2 = -1$?

Solution. We starting by finding the continued fraction expansion of $\sqrt{14}$:

$$\begin{aligned}
\sqrt{14} &= 3 + \sqrt{14} - 3 \\
\frac{1}{\sqrt{14}-3} &= \frac{\sqrt{14}+3}{(\sqrt{14}-3)(\sqrt{14}+3)} = \frac{\sqrt{14}+3}{5} = 1 + \frac{\sqrt{14}-2}{5} \\
\frac{5}{\sqrt{14}-2} &= \frac{5(\sqrt{14}+2)}{(\sqrt{14}-2)(\sqrt{14}+2)} = \frac{\sqrt{14}+2}{2} = 2 + \frac{\sqrt{14}-2}{2} \\
\frac{2}{\sqrt{14}-2} &= \frac{2(\sqrt{14}+2)}{(\sqrt{14}-2)(\sqrt{14}+2)} = \frac{\sqrt{14}+2}{5} = 1 + \frac{\sqrt{14}-3}{5} \\
\frac{5}{\sqrt{14}-3} &= \frac{5(\sqrt{14}+3)}{(\sqrt{14}-3)(\sqrt{14}+3)} = \sqrt{14}+3 = 6 + \sqrt{14} - 3.
\end{aligned}$$

So $\sqrt{14} = [3; \overline{1, 2, 1, 6}]$.

			3	1	2	1	6
x	0	1	3	4	11	15	
y	1	0	1	1	3	4	
$x^2 - 14y^2$			-5	2	-5	1	

So the functional solution to $x^2 - 14y^2 = 1$ is $(x, y) = (15, 4)$. Since no values of x, y for which $x^2 - 14y^2 = -1$ are found in the table above before that solution, $x^2 - 14y^2 = -1$ has no integer solutions. This also follows from the fact that $x^2 \equiv -1 \pmod 7$ has no solutions $x \in \mathbb{Z}$, since -1 N 7. Hence, the fundamental unit in $\mathbb{Z}[\sqrt{14}]$ is $15 + 4\sqrt{14}$, and $\pm(15 + 4\sqrt{14})^n$ $(n \in \mathbb{Z})$ give all the units in $\mathbb{Z}[\sqrt{14}]$.

7. (Triangular numbers and Pell's equations) A *triangular number* is a positive integer of the form

$$T_n = \frac{n(n+1)}{2}, \qquad n \geq 1.$$

Let k be a positive integer such that $2k$ is not a perfect square. Show that there are infinitely many triangular numbers T_n that are k times a perfect square.

Solution. We want to find infinitely many positive integer solutions (n, m) to $T_n = m^2$, namely,

$$\frac{n(n+1)}{2} = km^2.$$

Multiplying both sides by 8 and completing the square, we see that it is equivalent to

$$(2n+1)^2 - 1 = 8km^2.$$

We rewrite this as $(2n+1)^2 - 8km^2 = 1$.

Since $2k$ is not a square, we know that $8k = 2k \cdot 2^2$ is not a square either. Thus, Pell's equation $x^2 - 8ky^2 = 1$ has infinitely many integer solutions (x, y) with $x > 1$ and $y > 0$. Moreover, $x^2 = 1 + 8ky^2$ implies x is odd. For each such solution (x, y),

$$n = \frac{x-1}{2} > 0, \qquad m = y > 0$$

provides a positive integer solution to $(2n+1)^2 - 8km^2 = 1$. Therefore, there are infinitely many positive integers (n, m) such that $T_n = m^2$.

8. (Pell solutions as rational approximations) Let $d > 1$ be a nonsquare integer.

(1) Let $(x, y) = (a, b)$ be the fundamental solution to Pell's equation $x^2 - dy^2 = 1$, and put $\omega = a + b\sqrt{d}$. Then
$$\varepsilon_n = \omega^n = x_n + y_n\sqrt{d}, \quad n \in \mathbb{Z},$$
with ± 1 attached, give all the solutions. Establish the recurrence relations
$$x_{n+2} = 2ax_{n+1} - x_n \quad \text{and} \quad y_{n+2} = 2ay_{n+1} - y_n$$
for all $n \in \mathbb{Z}$.

Solution. Note that
$$\omega\overline{\omega} = N(\omega) = a^2 - db^2 = 1,$$
so $\omega^{-1} = \overline{\omega}$ and $\overline{\omega}^{-1} = \omega$. In addition, $\omega^n = x_n + y_n\sqrt{d}$ and $\overline{\omega}^n = x_n - y_n\sqrt{d}$ imply that
$$x_n = \frac{\omega^n + \overline{\omega}^n}{2} \quad \text{and} \quad y_n = \frac{\omega^n - \overline{\omega}^n}{2\sqrt{d}}.$$
It follows that
$$\begin{aligned} x_{n+2} + x_n &= \frac{\omega^{n+2} + \overline{\omega}^{n+2}}{2} + \frac{\omega^n + \overline{\omega}^n}{2} \\ &= \big(\omega + \omega^{-1}\big)\frac{\omega^{n+1}}{2} + \big(\overline{\omega} + \overline{\omega}^{-1}\big)\frac{\overline{\omega}^{n+1}}{2} \\ &= \big(\omega + \overline{\omega}\big)\frac{\omega^{n+1}}{2} + \big(\overline{\omega} + \omega\big)\frac{\overline{\omega}^{n+1}}{2} \\ &= \big(\omega + \overline{\omega}\big)\frac{\omega^{n+1} + \overline{\omega}^{n+1}}{2} \\ &= 2ax_{n+1}. \end{aligned}$$
Similarly,
$$\begin{aligned} y_{n+2} + y_n &= \frac{\omega^{n+2} - \overline{\omega}^{n+2}}{2\sqrt{d}} + \frac{\omega^n - \overline{\omega}^n}{2\sqrt{d}} \\ &= \big(\omega + \omega^{-1}\big)\frac{\omega^{n+1}}{2\sqrt{d}} - \big(\overline{\omega} + \overline{\omega}^{-1}\big)\frac{\overline{\omega}^{n+1}}{2\sqrt{d}} \\ &= \big(\omega + \overline{\omega}\big)\frac{\omega^{n+1}}{2\sqrt{d}} - \big(\overline{\omega} + \omega\big)\frac{\overline{\omega}^{n+1}}{2\sqrt{d}} \\ &= \big(\omega + \overline{\omega}\big)\frac{\omega^{n+1} - \overline{\omega}^{n+1}}{2\sqrt{d}} \\ &= 2ay_{n+1}. \end{aligned}$$

(2) Show that every solution $x, y > 0$ to Pell's equation $x^2 - dy^2 = 1$ satisfies
$$0 < \frac{x}{y} - \sqrt{d} < \frac{1}{2\sqrt{d}\,y^2}.$$
Consequently, since $x^2 - dy^2 = 1$ has infinitely many positive solutions, there are infinitely many rational numbers x/y, where $x, y > 0$ and $\gcd(x, y) = 1$, whose distance to $\sqrt{d}$ is less than $1/(2\sqrt{d}\,y^2)$. In particular, the sequence of these rational numbers, when arranged with increasing denominators, converges to $\sqrt{d}$.

Solution. Since $x, y > 0$, we have $x + y\sqrt{d} > 0$. Combining this with

$$(x + y\sqrt{d})(x - y\sqrt{d}) = 1,$$

we get $x - y\sqrt{d} > 0$, which implies $x > y\sqrt{d}$ and

$$\frac{x}{y} - \sqrt{d} > 0.$$

Moreover,

$$1 = (x + y\sqrt{d})(x - y\sqrt{d}) > (y\sqrt{d} + y\sqrt{d})(x - y\sqrt{d}) = 2y\sqrt{d}(x - y\sqrt{d}).$$

Dividing both sides by $2\sqrt{d}\,y^2$ yields

$$\frac{x}{y} - \sqrt{d} < \frac{1}{2\sqrt{d}\,y^2}.$$

(3) Based on the preceding parts, find a rational number x/y, where $x, y > 0$ and $\gcd(x, y) = 1$, such that

$$0 < \frac{x}{y} - \sqrt{3} < 10^{-6}.$$

You may use the fact that $(x, y) = (2, 1)$ is the fundamental solution to $x^2 - 3y^2 = 1$.

Solution. Let $\omega = 2 + \sqrt{3}$ and $\omega^n = x_n + y_n\sqrt{3}$. Then $(x_0, y_0) = (1, 0)$ and $(x_1, y_1) = (2, 1)$. Part (1) supplies the recurrence relations

$$x_{n+2} = 4x_{n+1} - x_n \quad \text{and} \quad y_{n+2} = 4y_{n+1} - y_n.$$

Using these we obtain

n	0	1	2	3	4	5	6
x_n	1	2	7	26	97	362	1351
y_n	0	1	4	15	56	209	780

By part (b), we need

$$\frac{1}{2\sqrt{3}\,y_n^2} \le 10^{-6},$$

i.e.,

$$y_n \ge 500\sqrt{\frac{2}{\sqrt{3}}}.$$

This certainly holds if $y_n \ge 500 \cdot 1.5 = 750$, since

$$\sqrt{\frac{2}{\sqrt{3}}} < \sqrt{2} < 1.5.$$

Our table above shows that $y_6 = 780 > 750$. Hence, $(x, y) = (x_6, y_6) = (1351, 780)$ satisfies

$$0 < \frac{x}{y} - \sqrt{3} < 10^{-6}.$$

Optional challenge

9. (A Diophantine equation) Find all integer solutions $(x, y) \in \mathbb{Z}^2$ to $x^2 + 2 = y^3$.

Solution. First, we show that $2 \nmid x$. If $2 \mid x$, then $2 \mid y$. So we may write $x = 2x_1$ and $y = 2y_1$. Substituting these into the equation, we get $4x_1^2 + 2 = 8y_1^2$, which is equivalent to $2x_1^2 + 1 = 4y_1^2$. This is impossible, since the left-hand side is odd while the right-hand side is even. Alternatively, we may argue that $2 \mid y$ implies that $4 \mid y^3 = y^2 \cdot y$, and so $x^2 + 2 \equiv 0 \pmod 4$, i.e., $x^2 \equiv 2 \pmod 4$, which is also impossible. Hence, $2 \nmid x$.

Working in the UFD $\mathbb{Z}[\sqrt{-2}]$, we have

$$x^2 + 2 = (x + \sqrt{-2})(x - \sqrt{-2}) = y^3.$$

We claim that $\gcd(x + \sqrt{-2}, x - \sqrt{-2}) = 1$. To see this, assume that some prime $\pi \mid \gcd(x + \sqrt{-2}, x - \sqrt{-2})$. Then

$$\pi \mid (x + \sqrt{-2}) - (x - \sqrt{-2}) = 2\sqrt{-2} = (\sqrt{-2})^3.$$

Since $\sqrt{-2}$ is a prime and the units in $\mathbb{Z}[\sqrt{-2}]$ are ± 1, we know that $\pi = \pm\sqrt{-2}$. Together with $\pi \mid x + \sqrt{-2}$, this implies $\pi \mid x$. Taking the norms, we see that $N(\pi) = 2 \mid N(x) = x^2$, which implies $2 \mid x$, a contradiction. So $\gcd(x + \sqrt{-2}, x - \sqrt{-2}) = 1$.

Consequently, there are $a, b \in \mathbb{Z}$ such that

$$x + \sqrt{-2} = (a + b\sqrt{-2})^3 = (a^3 - 6ab^2) + (3a^2b - 2b^3)\sqrt{-2}.$$

Comparing coordinates, we find

$$x = a^3 - 6ab^2, \qquad 1 = 3a^2b - 2b^3 = b(3a^2 - 2b^2).$$

The second equation implies

$$b = 1, 3a^2 - 2b^2 = 1 \qquad \text{or} \qquad b = -1, 3a^2 - 2b^2 = -1.$$

The first scenario gives $(a, b) = (\pm 1, 1)$, while the second yields $3a^2 = 2b^2 - 1 = 1$, impossible.

Plugging $(a, b) = (\pm 1, 1)$ in $x = a^3 - 6ab^2$, we conclude $x = \pm 5$. Finally, plugging $x = \pm 5$ in $x^2 + 2 = y^3$, we deduce $y = 3$. Therefore, the only integer solutions (x, y) to $x^2 + 2 = y^3$ are $(\pm 5, 3)$.